Chang Zeng

Senior Research Scientist | Generative Audio & Multimodal AI

Tokyo, Japan | (+86)13821557429 | zengchang.elec@gmail.com

| github.com/zengchang233 | scholar.google.com/citations?user=gfGyn49j-MkC&hl=en

Senior Research Scientist in **Generative Audio and Voice LLMs** with **7+ years** of experience from research to production. I have led end-to-end development of **expressive TTS** systems and served as a main contributor to a **full-duplex speech** system, covering data curation, audio tokenization/codec design, large-scale multi-GPU training, and deployment. My work has delivered state-of-the-art quality–efficiency trade-offs (including sub-1 kbps codec performance) and top benchmark rankings in multilingual settings. I publish actively at ICLR/Interspeech/ICASSP and specialize in turning advanced methods (**GANs, diffusion, transformers**) into product-ready voice AI.

Education

National Institute of Informatics (SOKENDAI) **2020.10 - 2024.03.22**

Doctor of Informatics

Sponsored by the Japanese government MEXT scholarship, supervised by Prof. Yamagishi.

The University of Tokyo **2017.10 - 2020.03**

Master of Electrical Engineering and Information System

Supervised by Prof. Minematsu.

Tianjin University **2012.09 - 2016.07**

Bachelor of Engineering

Publications

Featured Publications

1. **Zeng, C.**, & Wang, C. (2024). InstructSing: High-fidelity Singing Voice Generation via Instructing Yourself. *IEEE SLT 2024*. Demo page.
2. **Zeng, C.**, Miao, X., Wang, X., Cooper, E., & Yamagishi, J. (2024). Joint Speaker Encoder and Neural Back-end Model for Fully End-to-End Automatic Speaker Verification with Multiple Enrollment Utterances. *Journal of Computer Speech & Language. (JCR Q1)*
3. **Zeng, C.**, Wang, X., Cooper, E., Miao, X., & Yamagishi, J. (2022). Attention back-end for automatic speaker verification with multiple enrollment utterances. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 6717-6721). Project page. *(CoreRank B)*
4. Wang, C., **Zeng, C (co-first author)**., & He, X. (2023). Xiaoicesing 2: A High-Fidelity Singing Voice Synthesizer Based on Generative Adversarial Network. In *Proc. Interspeech 2023, 5401-5405*. Project page. *(CoreRank A)*
5. Wang, C., **Zeng, C (Co-first author)**., Zhang, B., Ma, Z., Zhu, Y., Cai, Z., Zhao, J., Jiang, Z., & Chen, Y. (2024). HAM-TTS: Hierarchical Acoustic Modeling for Token-Based Zero-Shot Text-to-Speech with Model and Data Scaling. Demo Page.
6. Li, K., Sang, W., **Zeng, C.**, Yang, R., Chen, G., Hu, X. (2024). LibriSpace: A Simulated Moving Audio Dataset for Speech Enhancement and Separation. *ICLR 2025.*

Full list: Google Scholar: https://scholar.google.com/citations?user=gfGyn49j-MkC&hl=en

Work Experiences

Shanda AI Research Tokyo **2025.09 - Present**

Senior AI Researcher

Responsibilities

- Lead R&D of expressive, high-quality TTS and full-stack voice-agent technologies for avatar and game products.

Achievements

- Lead a team of **2 full-time researchers + 1 intern** to build **KodamaTTS** from scratch on a Qwen-based foundation model for virtual human, world-model, and gaming applications.
- Build an end-to-end pipeline covering **data curation, codec design/training, multi-node multi-GPU LLM training, and evaluation**.
- Achieve **SOTA-level codec performance under 1 kbps**, surpassing strong objective baselines (e.g., PESQ).
- Reach **#2 ranking** across **3 languages** and **2 evaluation metrics** against major Chinese tech competitors.
- Serve as **Co-PI** in a joint project with Tsinghua University on cocktail-party speech interaction and deliver an initial **LLM-based universal data pipeline (HIVE)** and model architecture for audio separation.

Li Auto **2024.04 - 2025.09**

Multimodal Generative Researcher

Responsibilities

- Developed voice generation systems for Li Auto smart-space products and contributed to the multimodal foundation model **MindGPT-4o**.

Achievements

- Proposed **GFSQ (Group Finite Scalar Vector Quantization)** tokenizer for GPT-SoVITS, improving codebook utilization and decoding quality via delay-pattern decoding.
- Trained a **multi-timbre, multi-style** voice generation model for in-car voice-blog scenarios, enabling high-quality style/voice selection in production use cases.
- Built synthetic-data workflows with internal foundation TTS models to scale data diversity (accent, dialect, language, emotion, scenario).

- Led data production and **audio-head pre-/post-training** for a full-duplex conversational model, improving customization capability and robustness of speech interaction.

Xiaoice **2022.07 - 2023.07**

Avatar Research Intern

Responsibilities

- Focused on high-fidelity (48kHz) singing voice generation in collaboration with research and engineering teams.

Achievements

- Upgraded FastSpeech2-based XiaoiceSing to **XiaoiceSing2** using adversarial training; achieved near-human MOS (**4.23 vs 4.27**).
- Developed **HiFi-WaveGAN** vocoder with a novel pulse-sequence design, outperforming HiFi-GAN on 48kHz singing synthesis MOS (**4.23 vs 3.89**).
- Built **CrossSinger** (cross-lingual, multi-singer SVS in EN/JP/CN) with language-aware encoding and a bias eliminator; accepted at **ASRU 2023**.
- Improved training efficiency with **InstructSing** (DDSP + adversarial + UNet refinement), reducing convergence to **~20k steps** (about **10× faster** than common neural vocoders).
- Explored hierarchical acoustic modeling for token-based voice LMs (e.g., VALL-E style), improving pronunciation consistency.

ALIBABA **2020.04 - 2020.09**

Speech Recognition Researcher (Full-time)

Responsibilities

- Developed speech AI systems for Taobao Live compliance and broadcaster-risk control.

Achievements

- Built a large-scale **speaker recognition** system to verify broadcaster identity at scale in livestream scenarios.
- Developed a **spoken-term detection** pipeline for policy-sensitive/illegal word monitoring in live broadcasts.
- Researched SSL-based speech representations and implemented an end-to-end ASR system with knowledge distillation (ESPnet-based) for improved accuracy-efficiency tradeoff.

Activities

Competitions

- 4th/77 place for VoxCeleb Speaker Recognition Challenge 2019.
- 2nd/110 place for Zhijiang Cup Speech Recognition for Conversational Scenario 2021.
- 4th/42 place for Audio Deep Synthesis Detection Challenge 2022 track1 (Low-quality Fake Audio Detection, LF).
- 5th/27 place for Audio Deep Synthesis Detection Challenge 2022 track2 (Partially Fake Audio Detection, PF).

Academic Activities

- Research assistant at Yamagishi Lab, National Institute of Informatics
- ICASSP 2022 Oral Presentation.
- Interspeech 2022 Oral Presentation.
- Interspeech 2023 Poster Presentation.
- Reviewer of IEEE Open Journal of Signal Processing.
- Reviewer of ICASSP, ICME, Interspeech, NeurIPS, ICLR, ICML, and AISTATS.
- Organization committee of Joint Workshop of VoicePersonae and ASVspoof 2023.

Open Source

- WeSpeaker contributor (1.2k stars), ASV-Subtools contributor (636 stars), HIVE dataset (227 stars, 10k downloads on HF)

Skills

Tools and Languages	Python, CPP, Shell, Git, MySQL
Toolkits	PyTorch, Pytorch-Lightning, SpeechBrain, WeNet, WeSpeaker, Kaldi, Espnet, HuggingFace
Communication	Chinese (native), English (business level), Japanese (N2-125)